

AWS Cloud Computing

Lecture 1 Notes:

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Traditional Data Center (DC):



Customers have to invest in Capital expenses (CapEx) like land, Real-estate, Servers, Resources, Contracts, etc. along with the Operational expenses (OpEx) when considering On-premise/traditional data centers.

What is Cloud?

WHAT IS CLOUD?

**On-demand delivery of IT resources
through the **Internet** with
pay-as-you go pricing**

On-Demand IT Resources

Anywhere through Internet

Pay as You go Pricing

Cloud gives you flexibility to trade your CapEx with OpEx.

Advantages of Cloud:



Agility - The cloud gives you easy access to a broad range of technologies so that you can innovate faster and build nearly anything that you can imagine.

Elasticity - With cloud computing, you do not have to over-provision resources up front to handle peak levels of business activity in the future

Cost Savings - The cloud allows you to trade capital expenses (such as data centers and physical servers) for variable expenses, and only pay for IT as you consume it.

Deploy Globally - With the cloud, you can expand to new geographic regions and deploy globally in minutes.

IaaS vs PaaS vs SaaS:

Infrastructure (as a Service)	Platform (as a Service)	Software (as a Service)
Application	Application	Application
Data	Data	Data
Runtime	Runtime	Runtime
Middleware	Middleware	Middleware
Operating System	Operating System	Operating System
Virtualization	Virtualization	Virtualization
Servers	Servers	Servers
Storage	Storage	Storage
Networking	Networking	Networking

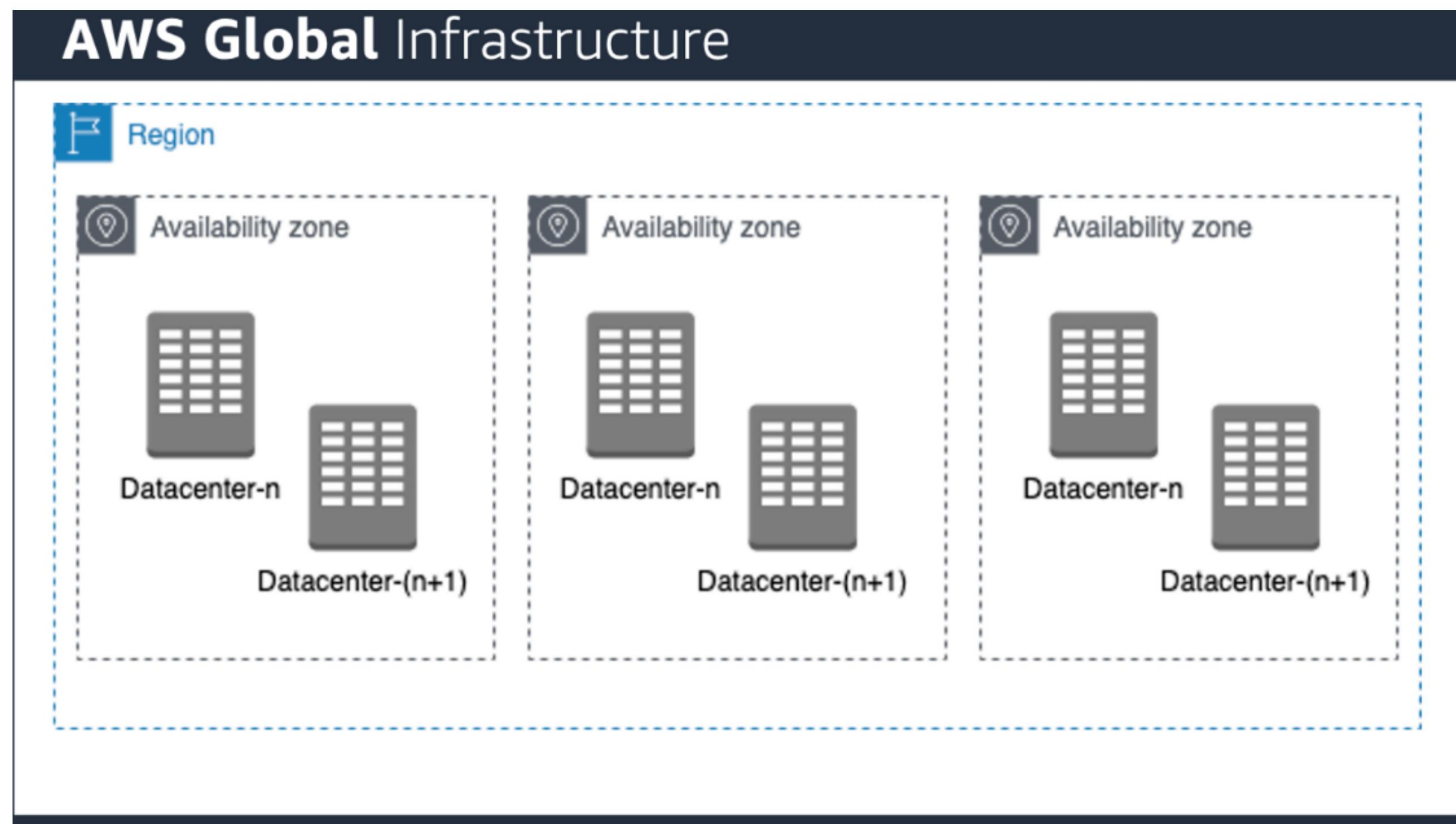
Examples:

IaaS – EC2, EBS, VPC

PaaS – Lambda, RDS, Elasticbeanstalk

SaaS – Gmail, Gdrive, O365

AWS Global Infrastructure:



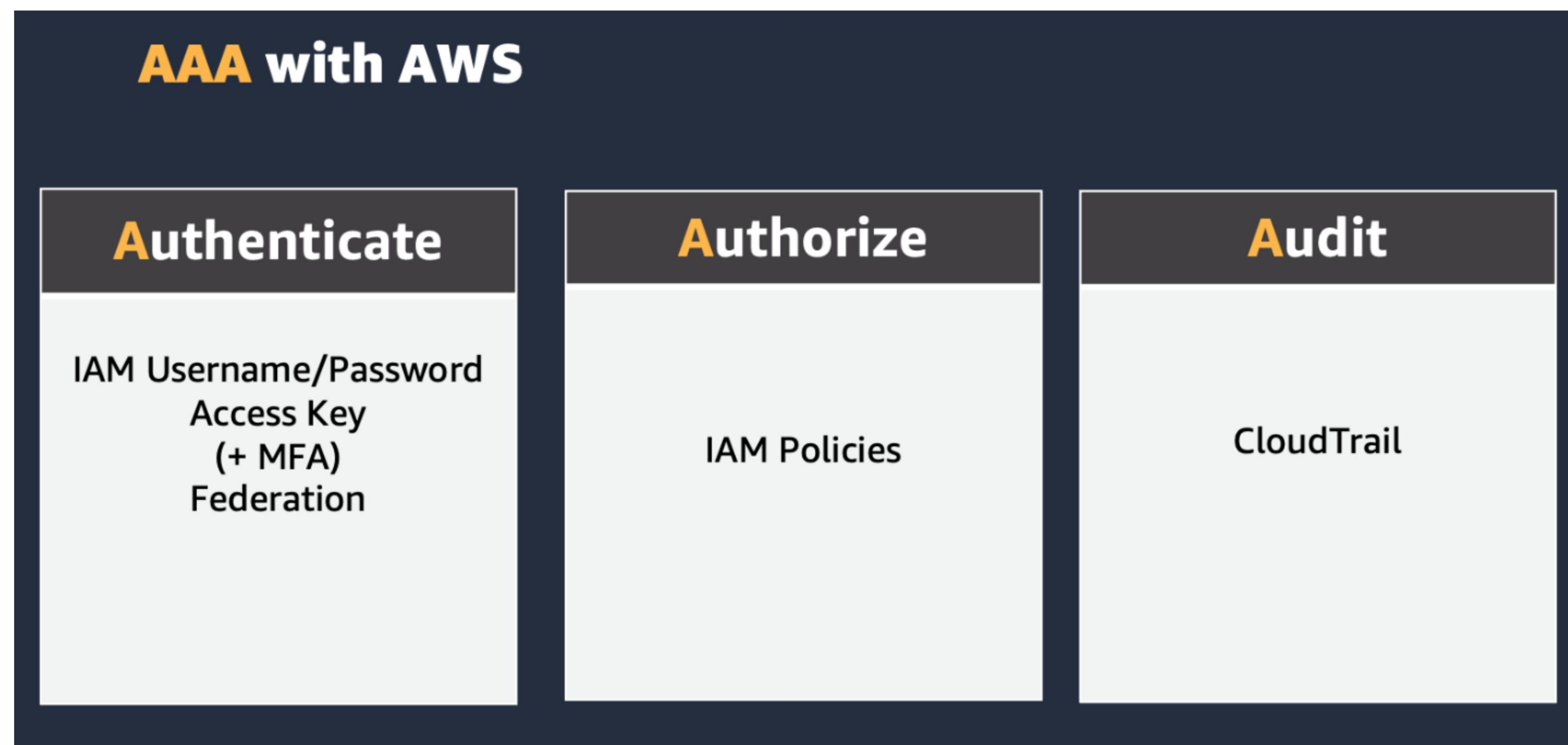
Regions

AWS has the concept of a Region, which is a physical location around the world where we cluster data centers. Each AWS Region consists of a minimum of three, isolated, and physically separate AZs within a geographic area.

Availability Zones

We call each group of logical data centers an Availability Zone. Each AZ has independent power, cooling, and physical security and is connected via redundant, ultra-low-latency networks. AWS customers focused on high availability can design their applications to run in multiple AZs to achieve even greater fault-tolerance. AWS infrastructure Regions meet the highest levels of security, compliance, and data protection.

IAM – Identity Access Management



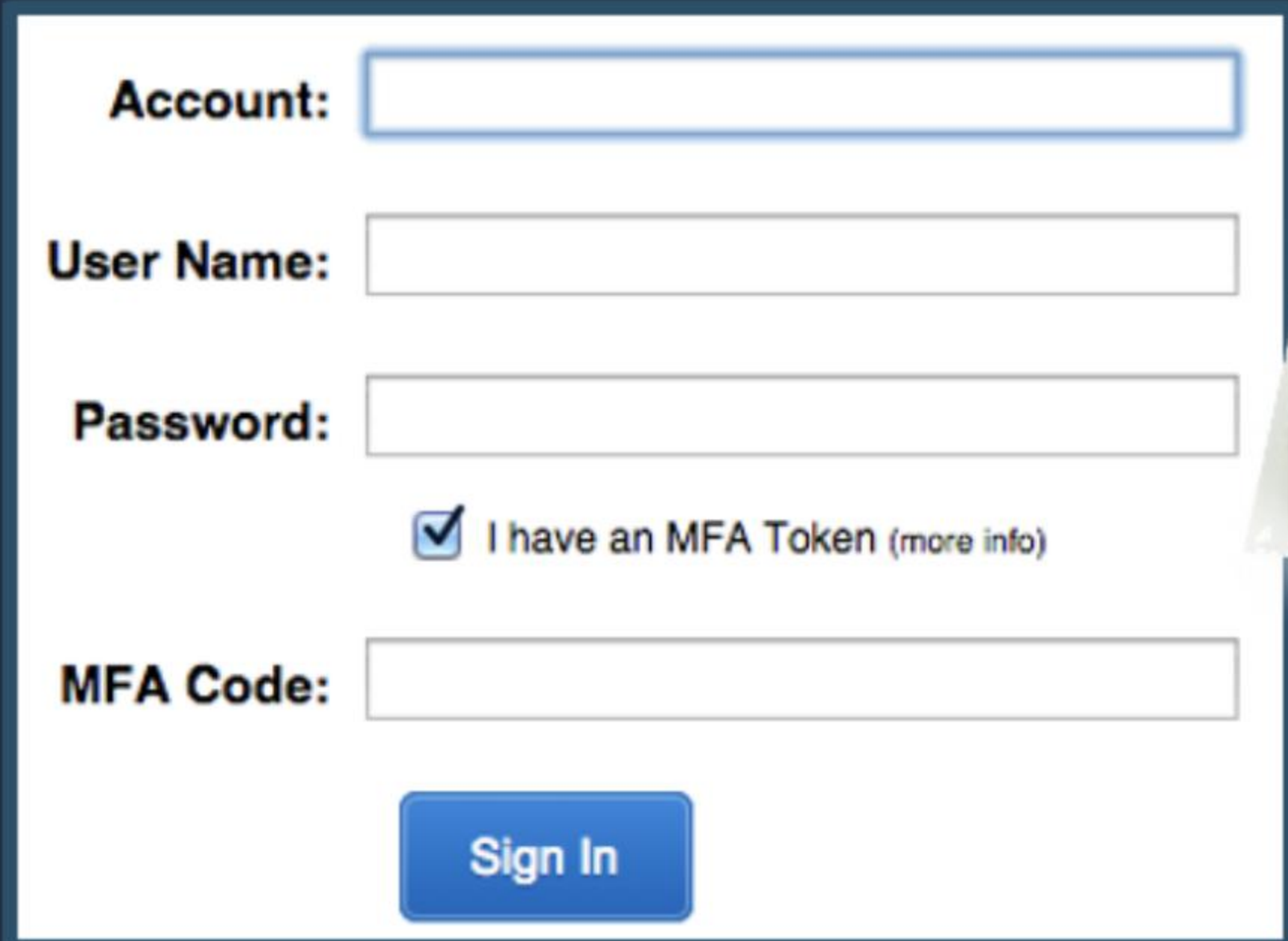
Authentication

AWS Identity Authentication

AUTHENTICATION: HOW DO WE KNOW YOU ARE WHO YOU SAY YOU ARE?

AWS Management Console

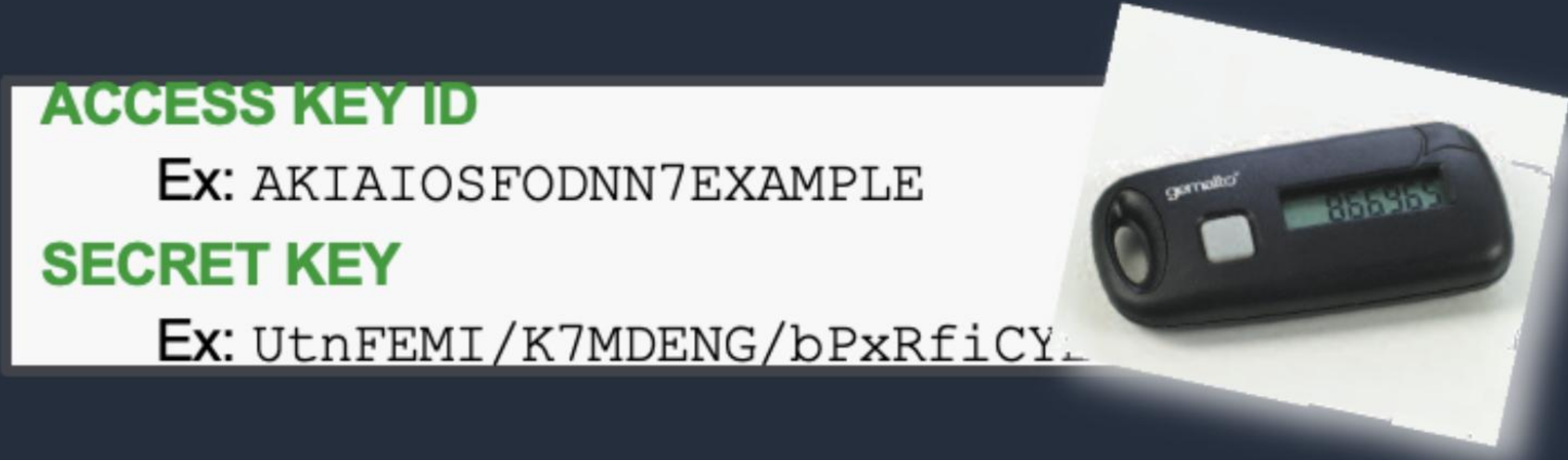
Login with **Username/Password** with optional **MFA** (recommended)



For time-limited access: a **Signed URL** can provide temporary access to the Console

API access

Access API using **Access Key + Secret Key**, with optional MFA



For time-limited access: Call the AWS Security Token Service (STS) to get a temporary AccessKeyId + SecretAccessKey + SessionToken

Authorization

AWS Authorization and Privileges

AUTHORIZATION: WHAT ARE YOU ALLOWED TO DO?

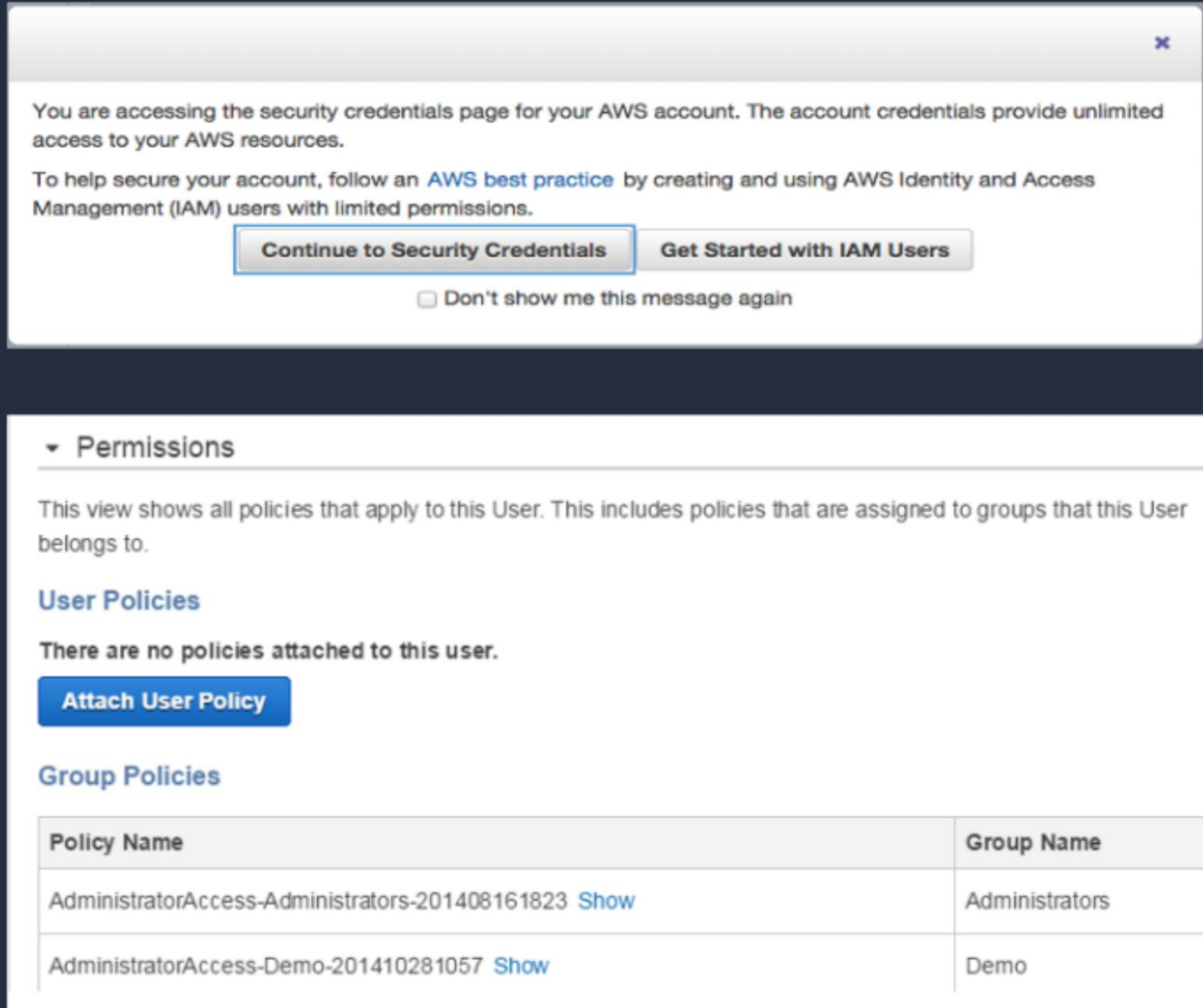
Account Owner (Root)

- Privileged for all actions.

Note: Always associate the account owner ID with an MFA device and store it in a secured place!

IAM Policies

- Privileges defined at User and Resource Level



Permissions

You are accessing the security credentials page for your AWS account. The account credentials provide unlimited access to your AWS resources.

To help secure your account, follow an [AWS best practice](#) by creating and using AWS Identity and Access Management (IAM) users with limited permissions.

[Continue to Security Credentials](#) [Get Started with IAM Users](#)

☐ Don't show me this message again

This view shows all policies that apply to this User. This includes policies that are assigned to groups that this User belongs to.

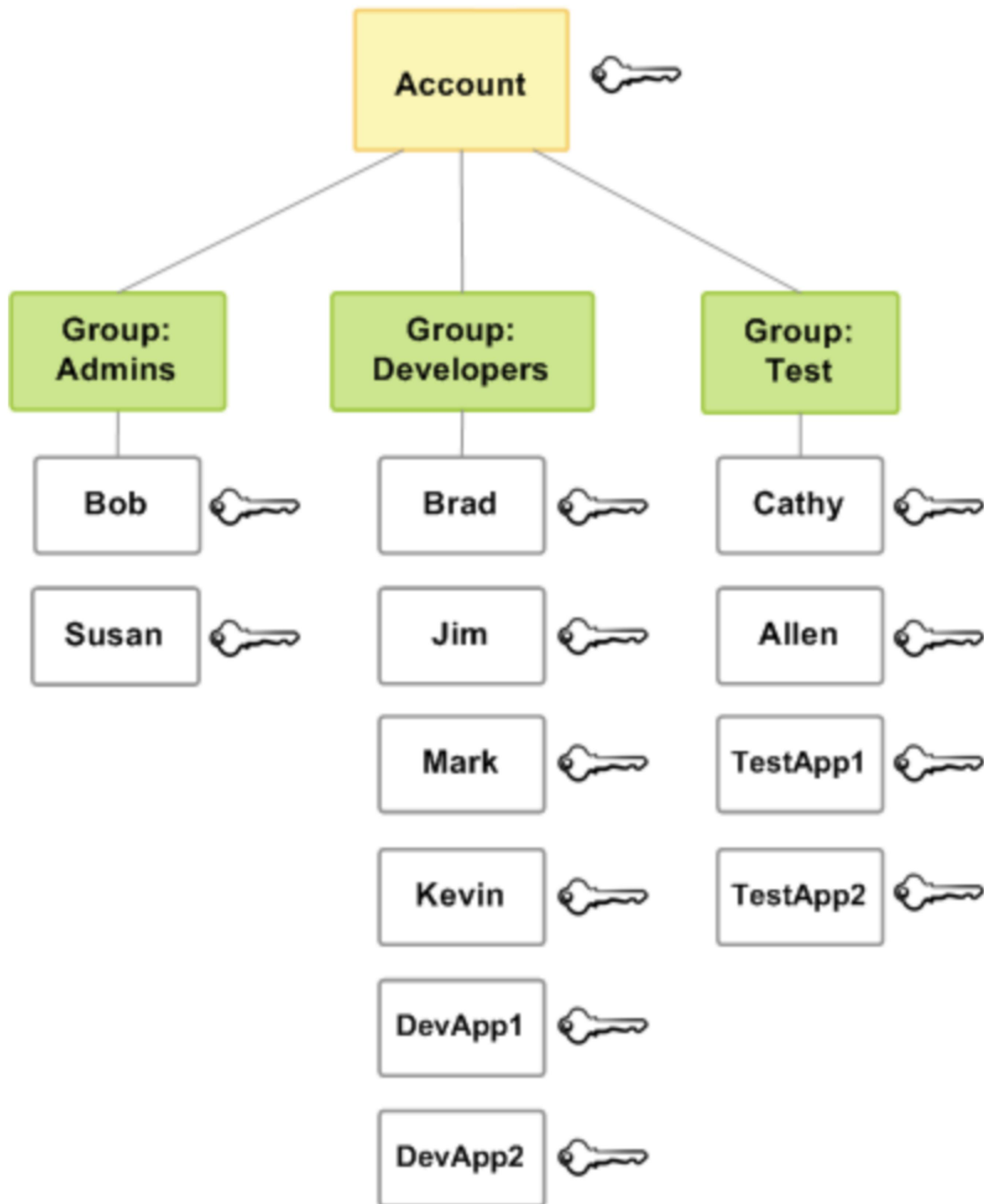
User Policies

There are no policies attached to this user.

[Attach User Policy](#)

Group Policies

Policy Name	Group Name
AdministratorAccess-Administrators-201408161823 Show	Administrators
AdministratorAccess-Demo-201410281057 Show	Demo



User Groups:

An IAM user group is a collection of IAM users. User groups let you specify permissions for multiple users, which can make it easier to manage the permissions for those users. For example, you could have a user group called *Admins* and give that user group typical administrator permissions. Any user in that user group automatically has *Admins* group permissions. If a new user joins your organization and needs administrator privileges you can assign the appropriate permissions by adding the user to the *Admins* user group. If a person changes jobs in your organization, instead of editing that user's permissions you can remove them from the old IAM groups and add them to the appropriate new IAM groups.

Users:

An *IAM user* is an entity that you create in your AWS account. The IAM user represents the human user or workload who uses the IAM user to interact with AWS resources. An IAM user consists of a name and credentials.

Policies:

A policy is an object in AWS that, when associated with an identity or resource, defines their permissions. AWS evaluates these policies when an IAM principal (user or role) makes a request. Permissions in the policies determine whether the request is allowed or denied. Most policies are stored in AWS as JSON documents.

Task 1

Create a DB group, attach policies and add user to it. Login to verify the required access has been applied to the user

Create user Groups

1. Go to AWS Console, search IAM
2. Click on User Groups and give a name “DBUserGroup”
3. Attach policies to it – RDSReadOnlyAccess
4. Click Next and create group

Create Users:

1. Click on Users on the left navigation pane
2. Give a username and password
3. Add the user to an existing group
4. Click Next and Create user
5. Save the Credentials

Login and Verify

Login with the new credentials and verify the access

Task 2

Create an Admin User, attach policies to it. Login to verify the required access has been applied to the user. Create and S3 bucket. Create access keys. Copy the file from local laptop to S3 using AWS Cli commands.

Create Users:

1. Click on Users on the left navigation pane
2. Give a username and password
3. Attach : AdministratorFullAccess policy
4. Click Next and Create user
5. Save the Credentials

Login and Verify

Login with the new credentials and verify the access

Create S3 bucket

1. Search for S3 in the AWS console
2. Click on Create bucket
3. Give an universally unique name to the bucket
4. Click on create.

Create Access Keys

1. Navigate to IAM > Click on User in the left navigation pane
click on your Admin user > Click on Security credentials tab
2. Click on create Access Keys
3. Save your access keys for future use

On your Laptop

1. Download AWS CLI software:
<https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html>
2. Install it on your system
3. Navigate to Powershell on your windows laptop
type command: `aws --version` (This will verify if AWS has been installed properly.)
4. Type command: `aws configure`
5. Now paste the access key and press enter
6. Paste the secret access key and press enter
7. For region type : `us-east-1`
8. For output Format: press enter

Copy files from Local Laptop to S3 bucket using CLI Commands

Now Navigate to the directory where you have stored few text files

Type command: **`aws s3 cp ./test.txt s3://yourbucketname/`**

You will see a message that file has been uploaded

List of Resources:

Certification Paths: https://d1.awsstatic.com/training-and-certification/docs/AWS_certification_paths.pdf

AWS Certified Cloud Practitioner (CLF-C02) Exam Guide: https://d1.awsstatic.com/training-and-certification/docs-cloud-practitioner/AWS-Certified-Cloud-Practitioner_Exam-Guide.pdf

AWS Certified Solutions Architect - Associate (SAA-C03) Exam Guide: https://d1.awsstatic.com/training-and-certification/docs-sa-assoc/AWS-Certified-Solutions-Architect-Associate_Exam-Guide.pdf

AWS Global Infrastructure: https://aws.amazon.com/about-aws/global-infrastructure/regions_az/

Setting up Cli: <https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html>

AWS CLI Commands: <https://docs.aws.amazon.com/cli/latest/>